



VLSI Testing

積體電路測試

Logic Simulation

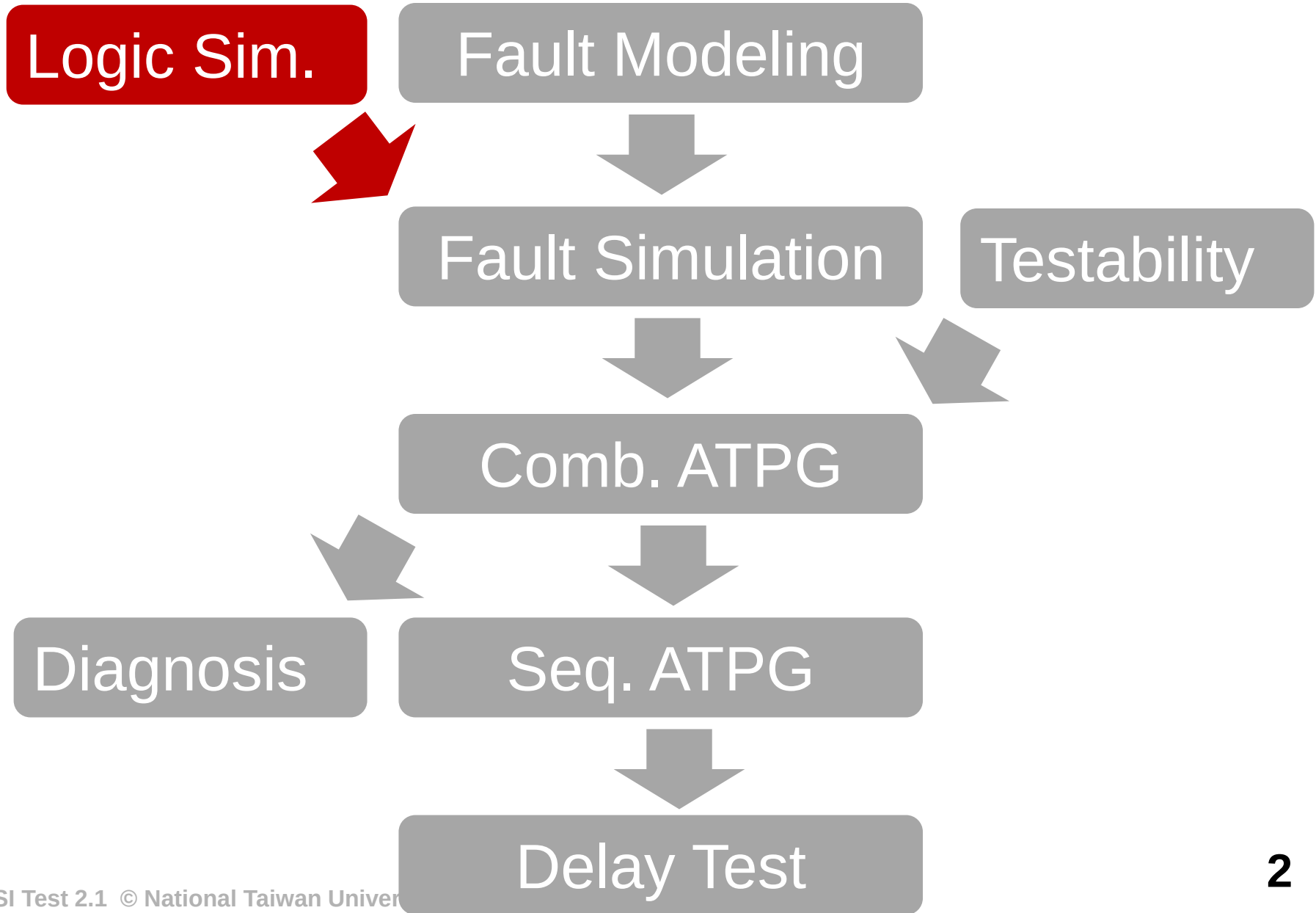
Professor James Chien-Mo Li 李建模

Lab. of Dependable Systems

Graduate Institute of Electronics Engineering

National Taiwan University

Course Roadmap (EDA Topics)



Why Am I Learning This?

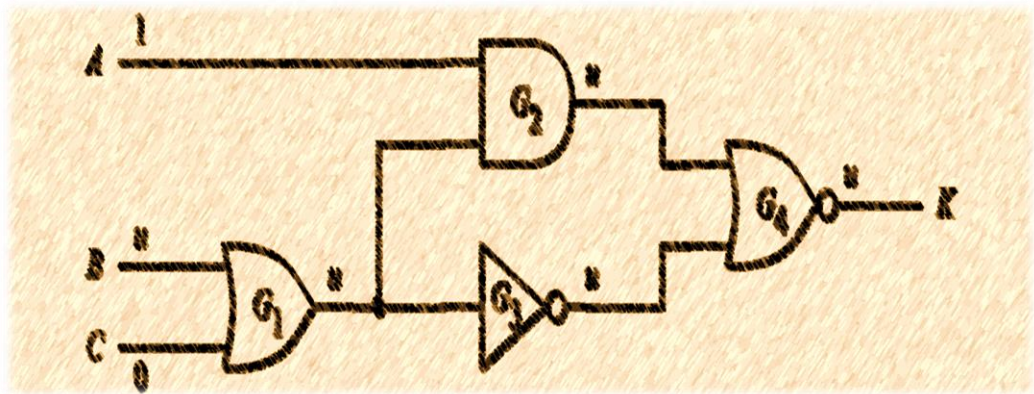
- Logic simulation is essential to verify our
 - ♦ Design correctness
 - ♦ ATPG patterns
- Logic simulation is basic EDA tool for digital circuits
 - ♦ Basic building block of many tools

“Logic is the anatomy of thought.”

(John Locke)

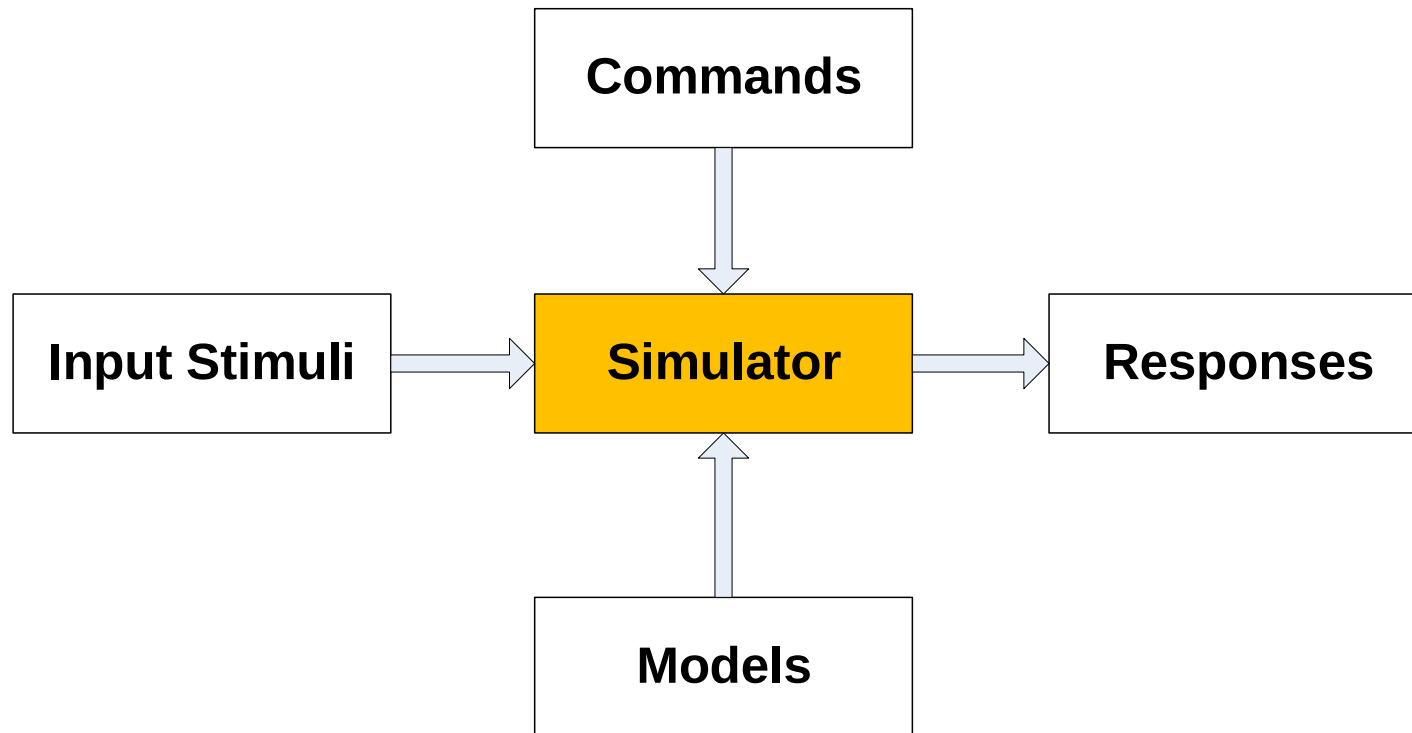
Logic Simulation

- Introduction
- Simulation Models
- Logic Simulation Techniques
- Issues of Logic Simulations
- Conclusions



What Is Simulation?

- Given input stimuli, models, and commands
 - ♦ Run software to produces output responses
- For digital circuits:
 - ♦ Input stimuli are *test patterns*
 - ♦ Models can be *functional/logic/transistors* (next slide)

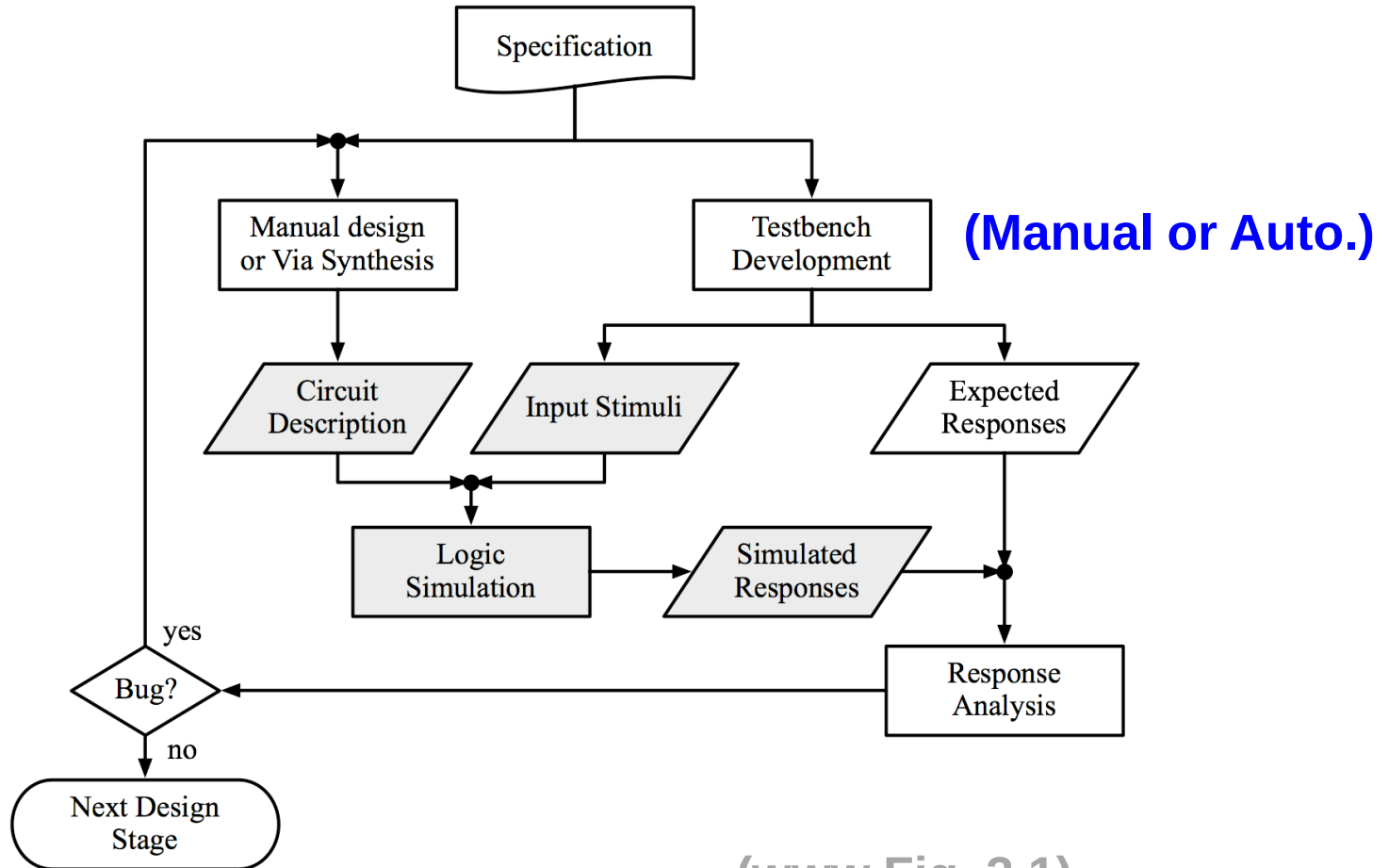


Levels of Circuit Simulation

- **Functional/Behavior level**
 - ♦ Interconnect of modules (expressed in programming language)
 - ♦ Good for architecture and functional verification
 - ♦ Fast but inaccurate
 - ♦ Example: C, System-C, Verilog (**Register-Transfer-Level, RTL**)
- **Logic level** ← focus of this chapter
 - ♦ Interconnect of Boolean gates (aka. **Gate-level netlist**): AND, OR, NOT ...
 - ♦ Good for logic verification, test pattern generation
 - ♦ Example: Verilog (gate-level)
- **Transistor level**
 - ♦ Interconnect of transistors
 - ♦ Good for analog/mixed-signal and timing critical digital circuits
 - ♦ Slow but accurate
 - ♦ Example: SPICE
- **Mixed-level**
 - ♦ Mixed functional/logic/transistor levels
 - ♦ Trade off CPU time and accuracy

Verification by Simulation

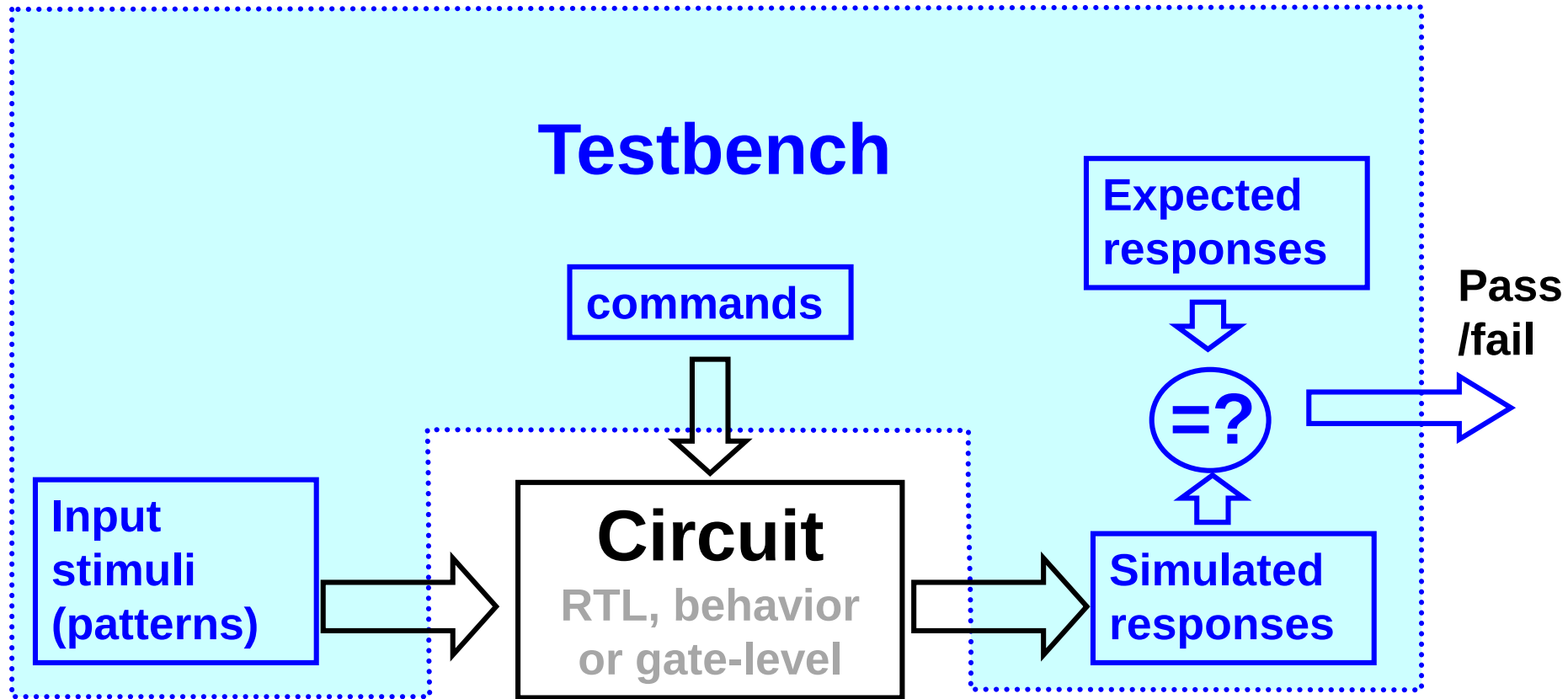
- 1. Verify design correctness
- 2. Verify ATPG patterns



(www Fig. 3.1)

What is Testbench ?

- Setup a simulation environment
- Apply input stimuli; run commands; observe responses



- Testbench is *NOT* implemented as physical circuits

Summary

- Introduction
 - ◆ What is simulation?
 - * Produce output based on given input stimuli and model
 - ◆ Levels of simulation
 - * functional, logic, transistor, mixed
 - ◆ Design verification by simulation
 - * testbench

